Journal of Ambient Intelligence and Humanized Computing (2024) 15:575–591
https://doi.org/10.1007/s12652-023-04715-5

ORIGINAL RESEARCH

EEG-based emotion recognition using modified covariance and ensemble classifiers

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Received: 22 March 2022 / Accepted: 25 September 2023 / Published online: 1 November 2023
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Abstract
The Electroencephalography (EEG)-based precise emotion identification is one of the most challenging tasks in pattern recognition. In this paper, an innovative EEG signal processing method is devised for an automated emotion identification. The Symlets-4 filters based "Multi Scale Principal Component Analysis" (MSPCA) is used to denoise and reduce the raw signal's dimension. Onward, the "Modified Covariance" (MCOV) is used as a feature extractor. In the classification step, the ensemble classifiers are used. The proposed method achieved 99.6% classification accuracy by using the ensemble of Bagging and Random Forest (RF). It confirms effectiveness of the devised method in EEG-based emotion recognition.

Keywords Electroencephalogram (EEG) · Emotion recognition · Modified Covariance (MCOV) · Ensemble machine learning

1 Introduction

The emotions are among the essential factors that can influence a person's behavior. To develop human behavior, it is necessary to understand and analyze their emotions. Understanding the human emotions and their consequences is critical for progress in the field of scientific computing. The emotional intelligence is required for human-computer interaction. The system will require a deep knowledge on human emotional experience to develop a successful identification of the emotional expressions. There are many things that affect human emotions such as watching movies, listening to songs and others. Therefore, the identification of these emotional changes can bring a significant advantage for the neuromarketing by understanding the consumer's choices (Daimi and Saha 2014). The Human-machine collaboration exists in many environments such as the industrial settings, offices, health, and others. As a result, researchers in the fields of scientific computing are studying the possibilities to increase the efficiency and flexibility of computer-human interaction. Additionally, they are trying to secure the high levels of user contentment. Therefore, in order to obtain a full grasp of human communication, the human computer interaction (HCI) systems must be able to adapt themselves to intended subjects. Human ideas can be shared verbally or nonverbally. The HCI systems aid in the understanding and analysis of the nonverbal human thinking. In particular while focusing on an automated understanding of a person's emotional responses, the process belongs to the domain of emotional computing (Yin et al. 2017).

The world is in rapid evolution, and many people disagree in expressing their feelings based on their cultural, social environment, age, and life experience. The need for sensors for human emotions understanding is becoming significant in many areas such as healthcare. In this case, a precise understanding of emotions can render a suitable treatment of several neurological and psychological diseases. The security field also needs these sensors in cases of criminal investigation, because sometimes the criminals can prevent feelings of fear and anxiety from appearing on their faces, but with these emotion sensors, it is difficult to circumvent the security team. Recently, there have been increasing demands for the development of these emotional sensors. Therefore,

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in our research we intend to reach the best modern technologies in human emotions recognition. Some researchers agree that data extracted from emotion sensors is inaccurate and needs to be developed. Sometimes it is hard to discriminate between feelings and emotions, because some feelings closely resemble like fear and anxiety. The electroencephalography (EEG) can be utilized to secure more accurate results, and to react quickly with human emotions (Tuncer et al. 2021b, p. 19), (Tuncer et al. 2021a). With the rapid development of technology, we can to take advantage of all new emotions sensors to improve the accuracy of extracted emotions to be more reliable in many potential applications (Tuncer et al. 2021b, p. 19), (Tuncer et al. 2021a).

1.1 Related works

The EEG signals processing for an automated emotions recognition is challenging. In this context, the domain adaption methods are a powerful tool for reducing the marginal distribution disparity (Chai et al. 2017). Because the EEG is a complex and fluctuating signal, consequently the characteristics retrieved from it varies drastically. However, the emotion states vary gradually and it can limit the performance of classifying the EEG based emotion data. However, the evolution of emotions over time is not demonstratable (Nandi et al. 2021). Because of the blurry borders, the recognizing of emotions from the EEG signals is challenging (Peng et al. 2021).

Mert and Akan, (2018), used the multivariate synchro squeezing transform (MSST) as a feature extraction technique. The "Independent Component Analysis" (ICA) was used to reduce the dimensionality of a 2D Time–Frequency (TF) distribution based presentation. The Artificial Neural Network (ANN) classifier was used to process the MSST-ICA based selected feature set. The system attained the accuracy scores of 82.11% and 82.03% respectively for the identification of arousal and valence states while processing the DEAP dataset.

Nakisa et al., (2018), suggested a novel framework that uses the "Evolutionary Computing" (EC) methods to automatically search for the best subset of the EEG characteristics. The proposed framework was tested using two publicly available datasets namely, MAHNOB and DEAP. Moreover, a novel dataset collected with the mobile EEG sensors was also studied. The findings show that the EC algorithms can successfully carried out the channels selection and features mining for maximizing the performance in a four-quadrant emotion classification task.

Khare et al., (2020, b), employed an adaptive "Tunable Q Wavelet Transform" (TQWT) for decomposing the EEG signals. The optimum configuration of the TQWT parameters was attained using the gray wolf optimization (GWO). The obtained sub-bands were analyzed for feature mining, which were onward processed by the least-squares support vector machine classifier. The authors reported a highest categorization accuracy of 95.70%.

Khare et al., (2020, b), used S-transform to attain the "Time-Order Representations" (TOR) of the EEG signals. Onward the TORs were processed by the CNN for the identification of human emotions. The highest accuracy score of 94.58% was reported.

Kim and Kang, (2018), looked into brainwaves and deep learning to see how much time people spend on their phones based on their emotions. The asymmetrical power of theta, alpha, beta, gamma, and total brainwave activity was assessed in 11 lobes. The deep belief network (DBN) was used as the deep learning technique together with "k-nearest neighbor" (k-NN) and a "Support Vector Machine" (SVM) to quantify the extent of smartphone addiction. The performance is tested on an EEG dataset, comprises of recordings from 25 subjects who watched videos expressing relaxation, fear, pleasure, and sadness. 13 subjects were from the non-risk group and 12 subjects were from the risk group. The model performed better for the emotion identification of non-risk group with an average accuracy score of 87.25%.

Al Zoubi et al., (2018), used the "Liquid State Machine" (LSM) to detect an individual's emotional state based on the EEG signals processing. The intended subjects watched emotional video snippets and were then judged based on their level of emotion during each clip. In this study, three emotions namely the aggression, arousal, and liking were considered. The authors have also discussed that how to employ the LSM's separation property to create a versatile emotion identification framework. They employed a single trained model to predict violence, arousal, and liking levels at various input durations. Tests are carried out using the DEAP dataset. The model attained more than 94% average accuracy score.

Kanjo et al., (2019), performed an iterative procedure that comprised adding and deleting a huge number of sensor inputs from various modes. They applied a deep learning approach for emotions classification. In a real-world scenario, the data came from smart phones and wearable gadgets. It creates a global model that depicts signal dynamics and temporal correlations by combining the local interactions of three sensor modes: environmental, on-body, and location. The model attained the accuracy of 83% for the considered dataset. It was concluded that the deep-learning approaches were effective in human emotion categorization when a large number of sensors were employed.

Liu et al., (2018), proposed an EEG feature extraction approach based on the "Empirical Mode Decomposition" (EMD) and "Approximation Entropy" (ApEn). The EMD was employed to treat EEG signals after data processing and produce numerous "Intrinsic Mode Functions" (IMFs). The ApEn of the first four IMFs was calculated and

utilized as a feature for learning and recognition from the EEG data. The suggested technique delivers an enhanced accuracy, which is extremely competitive with the state-of-the-art studies. The highest accuracy of 87.32%, while the average accuracy of 83.34% were secured.

Su et al., (2018), had suggested an ICA-based EEG signals spatial filtering approach. To find appropriate video stimuli for eliciting emotions, they first devised a stimulus selection approach based on the normalized violence/arousal space model. After that, by studying the ICA-to-electrode mapping patterns in various emotional states, an optimal ICA-based spatial filter was constructed. The developed filter then linearly projected on EEG data with five emotional conditions in terms of arousal and violence dimensions to extract attributes. Finally, they employed the SVM for classification. The results of the experiments proved that the suggested ICA-based spatial filtering method for emotional conditions identification is feasible. The highest accuracy score of 73.35% was secured for the valence emotion identification.

Katsigiannis and Ramzan, (2018), introduced the DREAMER, a multimodal database comprising of the EEG and Electrocardiogram (ECG) signals. These signals were captured during the affect elicitation by using an audio-visual stimuli. The signals of 23 participants were recorded after each stimulation, along with the self-assessment of their emotional states in terms of aggressiveness, arousal and dominance. All signals were captured using portable, wireless, low-cost, and off-the-shelf technology. These findings had pointed to the possibility of adopting the low-cost devices for emotion identification.

Zhao et al., (2018), had shown an individual's behavioral qualities by monitoring brain waves while analyzing the emotional content. Thirty-seven participants watched seven standardized video segments, targeting seven specific emotions, that described the real-life emotional experiences as a part of this study. When positive emotions were elicited, the model respectively accomplished the recognition performance for Extraversion of 81.08%, Agreeableness of 86.11%, and Conscientiousness of 80.56%.

Liu et al., (2018b), had created a standardized library of 16 emotive video snippets chosen from over 1,000 film samples. They suggested a real-time movie-generated emotion detection system for recognizing an individual's emotional conditions by measuring and processing the EEG signals. Thirty people took part in this study. Each subject watched 16 standardized film segments, targeting seven specific emotions as well as neutrality. These findings showed that EEG signals had an advantage over the reported state-of-the-art real-time emotion identification algorithms in terms of classification accuracy and the capacity to distinguish identical discrete emotions in the valence-arousal coordinate space.

Hwang et al., (2020), had extracted the topology-preserving differential entropy features. The jumping Hanning window is used to segment the EEG signals from the SEED dataset. Each segment contains 256 samples and onward the "Short-Time Fourier Transform" (STFT) is computed for each segment. The band-pass filters are used to divide the spectrum in five sub-bands. The differential entropy features are extracted from each sub-band. In next step, the polar projection is applied to attain the topology preserving representation of the mined differential entropy features. Finally the Convolutional Neural Network (CNN) is used for the classification while securing 90.41% accuracy score.

In (Zhong et al. 2020), the authors had employed the local and global inter-channel relation via adjacency matrix for features extraction from the SEED dataset. The selected feature set was conveyed to the "Regularized Graph Neural Network" (RGNN) for an automated categorization of emotions. This model secured the 94.24% accuracy score.

In (Wang et al. 2020), the authors had followed a similar sequence as presented in (Hwang et al., 2020) to obtain the sub-bands of short length segmented EEG signals from the SEED dataset. Onward, the Electrode-frequency distribution maps (EFDMs) were generated by normalizing the obtained sub-bands spectra. Finally, the CNN is sued for the classification purpose. This model achieved the 90.59% accuracy score.

Gao et al., had followed the scheme, presented in (Hwang et al., 2020), to extract the differential entropy features from the EEG signals (Gao et al. 2020). Onward, the spatial-temporal information is extracted from the differential entropy features. Afterward, the classification is carried out via the channel fused dense convolutional network (CDCN) model. This method secured 90.63% accuracy score while processing the SEED dataset.

Wu et al., had derived five sub-bands of the intended EEG signals by using the band-pass filters. Onward, they extracted the strength, clustering coefficient and Eigenvector centrality from considered sub-bands. These features were fused with the statistically mined eye movement features. The selected feature set was then classified by the Deep Canonical Correlation Analysis (DCCA) model while securing 95.08% accuracy score for the SEED dataset.

Tuncer et al., (2021b, p. 19), used the MSPCA for EEG signals denoising. Features ware extracted using the TQWT. S-box of the LED block cipher was utilized with statistical calculations based features mining from the TQWT generated sub-bands. The dimensionality is reduced using ReliefF and iterative Chi2 feature selectors. The model secured 94.58% accuracy for arousal for the case of DREAMER dataset and 99.29% accuracy for the case of GAMEEMO dataset.

Subasi et al., (2021), introduced a novel EEG-based automatic emotion identification system. A signal denoising

phase, a feature extraction phase, a dimension reduction phase, and a recognition phase are the four primary stages of the proposed technique. In the signal denoising phase, the MSPCA is employed to reduce noise level in the EEG signals. The feature extractor is a TQWT. For dimension reduction, six distinct statistical calculations were made. The "Rotation Forest Ensemble" (RFE) based classification was carried out. With RFE + SVM based ensemble, the suggested system achieved a more than 93% classification accuracy for the case of SEED dataset. The experimental findings had demonstrated that the suggested TQWT and RFE-based emotion detection framework was a successful method for recognizing emotions from the EEG data.

1.2 Contribution

The EEG signals frequently have low Signal to Noise Ratio (SNR) and considerable subject variability. Therefore, the single classifier models can perform poorly while processing the hand-crafted features. The intrinsic properties of EEG signals can be captured in a better way by employing effective denoising and features selection with the ensemble classifiers (Subasi et al. 2021).

This study presents the influence of movies on emotion formation. Moreover, numerous approaches for the EEG-based emotion classification by utilizing various ensemble models are investigated. The main contributions of this study are:

- We propose an original and effective hybridization of the Symlets-4 filter based MSPCA, Modified Covariance (MCOV), and ensemble learning for mining the neuronal patterns and distinguishing between the considered classes of emotions. The EEG signal denoising is achieved using the MSPCA. Onward, features are extracted by using the MCOV and then processed by the ensemble learning based models.
- The suggested processing chain is presented in this research as a way to accomplish the state-of-the-art emotional EEG data categorization. We secure a superior categorization accuracy while assuring a lower computational cost as compared to the previous related works.

To the best of the authors' knowledge, it is the first attempt to achieve an automated identification of emotions in the above enumerated manner. The findings are promising in terms of securing a relatively lower computational cost while achieving a higher emotion detection performance compared to the contemporary methods. To avoid any biasness in findings multiple categorization performance measures are considered namely, the Accuracy (Acc), F-measure (F1), Kappa statistics, and area under the "Receiver Operating Characteristics" (ROC) curve (AUC).

The paper is arranged as follows. The materials and methods are presented in Sect. 2. We provide information on the experimentation details in Sect. 3. Section 3 also discusses the findings, and the conclusion is made in Sect. 4.

2 Materials and methods

The proposed system contains several processing blocks, used in a sequential topology, as shown in Fig. 1. A description of different processing stages is given in the following.

2.1 The dataset

The publicly available SJTU Emotion EEG Dataset (SEED) is employed in this study (Zheng and Lu 2015). The dataset is recorded while watching emotional video clips and wearing a precise EEG wearable. The 15 Chinese cinema snippets (neutral, positive, and negative emotions) are considered, each is around 4 min long. The EEG signals of 15 subjects were recorded. Out of the intended 15 subjects, 7 are males and 8 are females. Their average age is 23.27-years with a standard deviation (STD) of 2.37-years. For the feedback, the subjects were instructed to write down their emotional responses, on the questionnaire, to every video segment immediately after watching it. Every subject was needed to complete the trials for three sessions in order to explore neuronal signatures and consistent patterns across sessions and persons. The interval between the two sessions was at least 1 week. The EEG data and facial videos were both recorded at the same time. The EEG signals were recorded from a 62-channel electrodes cap. The electrodes were positioned by using the 10–20 standard. The recordings were performed at a sampling rate of 1000 Hz. The important frequency content of an EEG signal for emotion identification is between [0, 75]Hz (Subasi et al. 2021), (Yang et al. 2020). As a result, the EEG signals are low pass filtered using a digital filter with a cut-off frequency of 75 Hz. After that these signals are decimated to achieve a sampling rate of 200 Hz. The MATLAB is used for this processing.

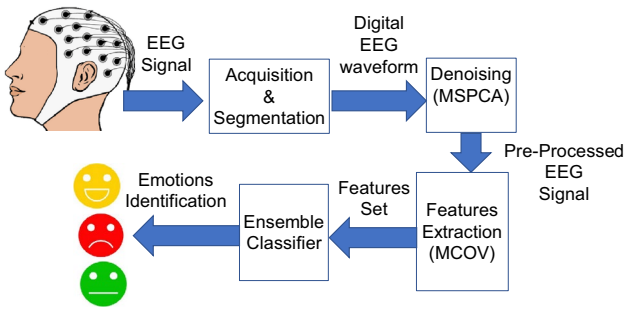


Fig. 1 The proposed EEG based Emotion Recognition method

2.2 Multi-scale principal component analysis (MSPCA)

The MSPCA combines the PCA's ability to extract deterministic features and eliminate nearly the relationship between auto-correlated measurements with wavelet analysis. The MSPCA calculates the PCA of the wavelet coefficients at each scale by integrating the findings at relevant scales. Because of its structure, the MSPCA is ideal for generating data that contains contributions from events whose behaviors change through time and frequency. The scales on which noteworthy events are discovered are constrained by MSPCA's process monitoring. Because of approximate de-correlation of wavelet coefficients, the MSPCA is also suitable for monitoring auto-correlated data without matrix augmentation or time-series modeling. Apart from improving the capacity to detect deterministic changes, MSPCA analysis also identifies the features that indicate improper procedure (Bakshi 1998), (Malinowski 2002).

The real error and the number of components that are maintained relates to the fixed error or the error which is maintained after erasing small principal components. It means that if only the principal components are not maintained, PCA can discard all of the errors. Sadly, the underlying relationship between variables, which beats the real goal of a latent variable regression method, will not be obtained by not keeping any parts. We can enrich the noise removal ability and the quality of the latent variable regression model if we combine it with wavelet thresholding. As noise relied on the relationship between variables is discarded by the former, wavelet thresholding of the latent variable and noise removal by getting rid of the latent variables that are of less importance supplement each other while noise relied on extracting features from measurements is discarded by the latter approach. The relationship between wavelet thresholding and latent variable models is made use of by present multi-scale empirical modeling methods (Bakshi 1998) (Alsberg et al. 1998) (Trygg and Wold 1998) (Bakshi et al. 1997).

In this study the Symlets-4 filters based wavelet analysis is used for decomposing the EEG signals. A 5th level of decomposition is used. The PCA of sub-band coefficients is performed at each scale to pick the principle components. Onward, the signal is reconstructed by wavelet synthesis. It provides a simplified and reduced noise-level version of the EEG signals. The MATLAB is used for implementing the MSPCA (Yang et al. 2020).

2.3 Feature extraction using the modified covariance

The modified covariance (MCOV) approach determines the parameters of the AR model by reducing the mean of the approximated error powers of the forward and the backward predictors (Bonat and Jørgensen 2016), (Seth et al. 2017). The forward and backward predictors are respectively given by the Eqs. (1) and (2).

$$\tilde{x}(n) = -\sum_{m=1}^{k} a(m)x(n-m). \tag{1}$$

$$\tilde{x}(n) = -\sum_{m=1}^{k} a^{*}(m)x(n+m). \tag{2}$$

The process of approximating the AR model parameters based on the procedure of decreasing the mean approximation error of forward and backward predictors is clear from the Eq. (3). Where, ρ_f and ρ_b can be expressed respectively via Eqs. (4) and (5). The minimization of the Eq. (3) can be realized by performing the complex gradient (Bonat and Jørgensen 2016), (Seth et al. 2017).

$$\rho = \frac{1}{2}\left(\rho_f + \rho_b\right). \tag{3}$$

$$\rho_f = \frac{1}{N-k}\sum_{n=k}^{N-1}\left|\left(x(n) + \sum_{m=1}^{k} a(m)x(n-m)\right)\right|^2. \tag{4}$$

$$\rho_b = \frac{1}{N-k}\sum_{n=0}^{N-1-k}\left|\left(x(n) + \sum_{m=1}^{k} a^{*}(m)x(n+m)\right)\right|^2. \tag{5}$$

The process is depicted, in matrix form, with the help of following Eq. (6).

$$\begin{bmatrix} c(1,0) \\ \cdot \\ \cdot \\ c(k,0) \end{bmatrix} + \begin{bmatrix} c(1,1) \ldots\ldots\ldots\ldots .c(1,k) \\ \cdot \\ \cdot \\ c(k,1) \ldots\ldots\ldots\ldots .c(k,k) \end{bmatrix} \begin{bmatrix} a(1) \\ \cdot \\ \cdot \\ a(k) \end{bmatrix} = \begin{bmatrix} 0 \\ \cdot \\ \cdot \\ 0 \end{bmatrix}. \tag{6}$$

Equation (6) can be symbolized as Eq. (7).

$$c_K + C_K a = 0. \tag{7}$$

In Eq. (6), c(j,m) is given by the Eq. (8).

$$c(j,m) = \frac{1}{2(N-k)}\left(\sum_{n=k}^{N-1} x^{*}(n-j)x(n-m) + \sum_{n=0}^{N-1-k} x(n+j)x^{*}(n+m)\right). \tag{8}$$

Parameters of the AR model can be determined from theEquation (9) by using the Eq. (7).

$$a = -c_k C_k^{-1}. \tag{9}$$

By utilizing the estimated AR model parameters, the power spectral density (PSD) could be calculated by using

the Eq. (10). Where, σ^2 presented the variance of white noise (Bonat and Jørgensen 2016), (Seth et al. 2017) (Subasi and Qaisar 2020).

$$P_{MCOV}(f) = \frac{\sigma^2}{\left|1 + \sum_{m=1}^{k} a(m)e^{-j2\pi fm}\right|^2}. \quad (10)$$

In this study, the optimal AR model is found with order of 15. 40 PSD coefficients are used to present each instance to get better performance.

2.4 Ensemble machine learning

The significance of ensemble modeling is that it allows to develop and integrate many single models for the same domain, potentially improving prediction quality for most or all of them. Individual models' strengths should be kept or reinforced, while their shortcomings should be removed or minimized, in order for this growth to be feasible. It turns out that a group of dozens or hundreds of models, even with low performance, may provide top-notch forecasts. Ensemble modeling may be used for both classification and regression, which are two of the most common predictive modeling tasks. Even if each individual model is entirely human readable, it may offer a significant gain over single models at the cost of significantly more computing time for various model development and a loss of overall human readability. Appropriate strategies for base model development and aggregation are necessary to fully harness this potential for improved predictive power (Cichosz 2014).

Our test harness gives us an indication of how well the model will perform on average, which is commonly assessed using repeated k-fold cross-validation as the gold standard. Moreover, using a variety of modeling algorithms or training data sets, ensemble modeling is the process of building numerous varied models to predict an outcome. The ensemble model then combines each base model's forecast into a single overall prediction for the unobserved data. The goal of employing ensemble models is to lower the prediction's generalization error. The ensemble approach reduces the model's prediction error as long as the base models are diverse and independent. The strategy looks to the collective wisdom of people to make a forecast. The ensemble model behaves and functions as a single model even when it has numerous basis models. Ensemble modeling is the practice of predicting a result using numerous diverse base models. The goal of employing ensemble models is to reduce forecast generalization error. When the ensemble approach is applied, the prediction error lowers as long as the basis models are diverse and independent, hence more reliable. Ensembles trained under the same conditions might have dramatically different robustness, particularly in terms of out-of-distribution detection capabilities. We see that, in terms of robustness measures, choosing ensemble members based on the discussed diversity metrics seldom exceeds the baseline of a selection based on accuracy across all configurations, employing different datasets and model architectures. We conclude that there is tremendous opportunity to improve the construction of robust ensemble models, and that novel and more complex diversity measurements could help. The variation in robustness among equally trained ensembles, and if diversity metrics are appropriate for indicating ensemble robustness. Our results reveal that ensembles trained under identical settings can differ greatly in terms of performance and robustness. Concerning the subject of whether diversity measures are appropriate indicators of ensemble robustness, we discovered that disagreement, normalized disagreement, and output correlation are not. A model's performance and predictions' spread, or dispersion are lessened by an ensemble. When a single predictive model is not enough to solve a problem involving predictive modeling, ensembles are used. This is accomplished by adding bias to the model, which reduces the variance component of the prediction error (i.e. in the context of the bias-variance trade-off). Another significant and underappreciated advantage of ensemble methods is improved robustness or reliability in a model's average performance (Tripathi et al. 2021) (Zhao et al. 2021) (Heidemann et al. 2021).

2.4.1 Bagging

Different models' decisions can be integrated in a single forecast. In the case of classification, taking a vote is the simplest method. Bagging gives equal weight to all models. However, the boosting weighting gives more weight to the more effective performers, evaluated on the basis of their past performance. In bagging, a decision tree is built for each dataset by randomly selecting many training datasets of the same size from the issue area. To produce the same forecast for each new test instance, these trees are almost similar. However, this is not always the case, especially when the training datasets are small. This invariably implies that in some test cases, some decision trees achieve precise classification while others do not. Bagging attempts to alleviate the instability of machine learning algorithms by randomly picking instances from the original dataset and replacing them with new ones of the same size. Some of the instances will be deleted and the remainder will be duplicated as a result of this sampling approach. Bagging employs each of these resultant datasets in the learning technique, and the outputs made from them vote for the class to be predicted. Bagging simply resamples the existing training data rather than generating independent datasets from the domain. Because they are all formed from the same dataset, the datasets created by resampling are not independent. Bagging, on the other hand,

produces a combined model that, on average, outperforms the single model constructed from the original training data (Witten et al. 2016).

Bootstrap samples ($\mathbf{T}_b$), with b = 1, 2,…, B, are obtained from the training set $\mathbf{T}_n = \{(\mathbf{x}_1, y_1),\ldots, (\mathbf{x}_n, y_n)\}$, starting with the training set where y_i takes values at 1, 2,…, k.By drawing with replacement, the same number of elements as in the initial set, these bootstrap replicates are obtained. The bagging method is specifically used as follows. Each $\mathbf{T}b$ replica builds a simple classifier $C_b(\mathbf{x}_i)$, where $C_b(\mathbf{x}_i)$ exactly like y_i accepts values at 1, 2,…, k. Then, using a majority vote, the B basic classifiers are combined, with the most frequently predicted class being selected as the final decision rule:

$$C_f(x_i) = \underset{j = \{1,2,\ldots,k\}}{\operatorname{argmax}} \sum_{b=1}^{B} \mathbf{I}\left(C_b(\mathbf{x}_i) = j\right) \tag{11}$$

The indicator function, $\mathbf{I}(i, j)$ produces 1 when the inner expression evaluates to true and 0 otherwise (Alfaro et al. 2018).

2.4.2 AdaBoost

It may be used to any classification learning problem, just like bagging. To make things easier, we assume that the learning algorithm can handle weighted samples, where the weight of each sample is a positive number. When sample weights are taken into account, a classifier's error is calculated as the sum of the mislabeled samples' weights divided by the total weight of all samples, rather than the proportion of mislabeled samples. By weighting samples, the learning algorithm may be driven to focus on a certain group of instances, particularly those with a high weight. The boosting process starts by giving all samples in the training data equal weight. The learning process is then used to create a classifier for this data, and each sample is reweighted based on the classifier's output. The weight of correctly categorized samples is reduced, while the weight of incorrectly classified samples increases. This results in a collection of "easy" occurrences with low weight and a collection of "hard" samples with high weight. A classifier is developed for the reweighted data in the following iteration—and all future ones—which concentrates on successfully categorizing the challenging occurrences. The weights of the samples are then increased or lowered based on the output of this new classifier. As a result, certain difficult situations may become even more difficult, and easier situations may become even easier. On the other hand, some difficult situations may turn out to be easier, and easier ones harder—all options are possible in practice. The weights represent how often the examples have been misclassified by the classifiers developed thus far after each iteration. This approach gives an elegant means of producing a sequence of experts that complement one another by retaining a measure of "hardness" with each occurrence. Boosting frequently results in classifiers, which are substantially more precise on new data than those produced by bagging. Boosting, unlike bagging, can, on occasion, fail in practical conditions. It can provide a classifier, which is far less accurate than a single classifier created from the same data. This implies that the data is overfitted by the combined classifier (Witten et al. 2016).

It can be assumed that there are just two classes without losing generality for the sake of simplification. $\mathbf{T}_n = \{(\mathbf{x}_1, y_1),\ldots, (\mathbf{x}_n, y_n)\}$, yields a training set, where yi has values of 1, 1. Each observation xi is given the weight w_b(i), which is originally set to 1/n. Each time a step is completed, this value is changed. This new training set is used to construct a basic classifier, labeled $\mathbf{C}_b(\mathbf{x}i)$, which is then applied to each training example. This classifier's error is denoted by the symbol b, and it is calculated as

$$\epsilon_b = \sum_{i=1}^{n} w_b(i)\mathbf{I}\left(C_b(\mathbf{x}_i) \neq y_i\right) \tag{12}$$

This procedure is performed for b = 1, 2, 3,…, B at each step. Finally, a linear combination of the single classifiers weighted by the appropriate constant α_b is used to create the ensemble classifier (Alfaro et al. 2018):

$$C_f(x_i) = sign \sum_{b=1}^{B} \mathbf{I}\left(C_b(\mathbf{x_i})\right) \tag{13}$$

2.4.3 MultiBoost

MultiBoost is a decision-making strategy, which combines AdaBoost with wagging, a type of Bagging, which is more appropriate for the job than straight Bagging. Experiments show that it is more successful at reducing errors in general than any of its constituent strategies. Wagging (Bauer and Kohavi 1999) is a version of bagging. It necessitates a basic learning algorithm, which may utilize training examples of varying weights. Wagging adds random weights to the samples in each training set rather than utilizing random bootstrap samples to create the succeeding training sets. The original wagging formulation by Bauer and Kohavi (1999) employed Gaussian noise to alter the instance weights. Some instance weights, however, may be decreased to zero, thereby eliminating them from the training set.

The findings that Bagging and AdaBoost appear to work through separate methods, have different impacts, and both have the largest influence on the first few committee members imply that combining the two may be beneficial. Because the processes differ, combining them may outperform either individually (Bauer and Kohavi 1999). Given that the Bagging decreases variance primarily, whereas AdaBoost decreases both bias and variance. It is evident that Bagging reduces variance more effectively than AdaBoost (Bauer and Kohavi 1999). Therefore, their combination

could be able to keep AdaBoost's bias decrease even though adding Bagging's variance decrease to what AdaBoost has already achieved. As a result, the MultiBoost may be seen as AdaBoost's wagging committees. In addition to the bias and variance reduction properties with each of its essential ensemble learning algorithms may inherit, MultiBoost has the latent computational advantage over AdaBoost in that sub-committees can be learned in parallel, though this would necessitate a change in how early termination of sub-committee learning is handled. Because the AdaBoost method is essentially sequential, it limits the amount of parallel processing that can be done. Each classifier learnt with wagging, on the other hand, is independent of the others, permitting parallel calculation, which MultiBoost receives at the sub-committee level (Webb 2000).

2.4.4 The random subspace method (RSM)

Ho (1998) suggested the RSM as an ensemble approach. In the RSM, the training data is also altered. However, this alteration is made in the feature space. The original p-dimensional feature space's r-dimensional random subspaces are produced. Consequently, the new training set now contains r-dimensional training objects. In the final decision rule, classifiers can be built in random subspaces and ensembled using the simple majority voting.

One can alter the training data in the RSM as well. This alteration is made in the feature space, though. In the training sample set $\boldsymbol{X} = (\boldsymbol{X}_1, \boldsymbol{X}_2, \ldots, \boldsymbol{X}_n)$, each training object $\boldsymbol{X}_i$ $(i = 1, \ldots, n)$ should be a p-dimensional vector $\boldsymbol{X}i = (x_{i1}, x_{i2}, \ldots, x_{ip})$, defined by p characteristics (components). In the RSM, $\boldsymbol{p}$-dimensional data set $\boldsymbol{X}$'s $\boldsymbol{r}$x$\boldsymbol{p}$ features are randomly chosen. Thus, one obtains the original $\boldsymbol{p}$-dimensional feature space's $\boldsymbol{r}$-dimensional random subspace. Consequently, the $\boldsymbol{r}$-dimensional training objects make up the modified training set. Then, in the final decision rule, one can create classifiers in the random subspaces and merge them by simple majority vote. As a result, the RSM is structured as follows:

(1) Repeat for $b = 1, 2, \ldots, B$, and so forth.

   (a) Select an $\boldsymbol{r}$-dimensional random subspace $\boldsymbol{X}\tilde{}\boldsymbol{b}$ from the original $\boldsymbol{p}$-dimensional feature space $\boldsymbol{X}$.
   (b) Create a classifier $C_b(\boldsymbol{x})$ (with a decision boundary $C_b(\boldsymbol{x}) = 0$) in $\boldsymbol{X}\tilde{}b$.

(2) Combine classifiers $C_b(\boldsymbol{x})$, $b = 1, 2, \ldots, B$, by simple majority voting to a final decision rule.

For both classifier development and aggregation, the RSM might benefit from using random subspaces. The small sample size problem may be overcome by creating classifiers on random subspaces. In fact, the number of training items is the same as in the original feature space, but the subspace dimensionality is lower. Consequently, the relative training sample rises in size. The random subspaces may provide stronger classifiers than the original feature space. It is particularly the case for datasets that contains several redundant attributes. The combined judgment of such classifiers may be superior to a single classifier, developed on the original training set (Skurichina and Duin, 2002).

2.4.5 Rotation forest (RoF)

The RoF is an ensemble learning approach that aims to produce varied yet accurate classifiers. It combines the random subspace and bagging techniques with the construction of main component features to produce an ensemble of decision trees. The input characteristics are randomly separated into k distinct subgroups in each cycle. The PCA is used to produce linear combinations of the qualities in each subset, which are rotations of the original axis. The derived characteristics are calculated using the k-sets of principal components that serve as the tree learner's input at each iteration. Because all of the components, gathered on each subset, are retained. Therefore, there are as many derived attributes as there are original attributes. To avoid the generation of identical attributes, if the same feature subset is used in numerous repeats, the PCA is conducted on the training instances from a randomly chosen subset of the class instances. To increase variation, a bootstrap sample of the data is created before the major component modifications are made in each iteration. Experiments reveal that rotation forests can perform similarly to "Random Forests" (RF), although having many fewer trees (Witten et al. 2016).

Suppose $x=[x_1; \ldots ; x_n]^T$ Let T be a data point with n features, and **X** be the training objects' data set in the form of a N x n matrix. Let $Y =[y_1; \ldots ; y_N]^T$ be a vector of class labels for the data, where y_j represents a value drawn from the set of class labels $\{w_1; \ldots ; w_c\}$. The ensemble's classifiers are denoted by $D_1; \ldots; D_L$, and the feature set is denoted by F. We have to choose L beforehand, as is typical for ensemble approaches. All classifiers can be trained simultaneously, and Bagging and Random Forests support this as well. We do the following actions to create the training set for classifier D_i:

(1) Randomly divide K subsets of F into (K is a parameter of the algorithm). The subsets could be intersecting or disjoint. We selected disjoint subsets to increase the likelihood of high variety. Let's assume for the sake of simplicity that K is a factor of n, resulting in each feature subset containing M = n/K features.
(2) Use $F_{i,j}$ to denote the jth subset of features for the classifier D_i's training set. Choose at random a nonempty subset of classes for each such subset, and then create

a bootstrap sample of objects that is 75% the size of the total number of data points. Run PCA using the chosen subset of **X** and only the **M** features in $F_{i,j}$. Store the coefficients of the principal components, each of size M x 1, as follows: $a_{i,j}^{1};\ldots;a_{i,j}^{M_j}$. Remember that some of the eigenvalues might be zero, which would mean that we might not have all M vectors and that $M_j \leq M$. In an effort to prevent similar coefficients if the same feature subset is chosen for different classifiers, PCA is run on a subset of classes as opposed to the entire set.

(3) Arrange the vectors with coefficients that were acquired in a sparse "rotation" matrix called R_i. We then rearrange the columns of R_i (the features) so that they correspond to the original features in order to generate the training set for classifier D_i. Indicate the new rotational matrix R_i^a (size N x n). So, XR_i^a is the training set for classifier D_i (Rodriguez et al. 2006).

3 Results

The main objective of EEG signal processing in emotion recognition is to establish a framework that allows to recognize the mood and mental status of a human. Figure 1 shows the framework of the suggested approach. After signal acquisition, the MSPCA is used to remove various forms of EEG noises and artefacts, as seen in this diagram. In the second stage, the pertinent attributes are retrieved to train the classifier by using the MCOV. When the classification process is done, the intended emotions are recognized.

3.1 Performance evaluation

The classification performance is evaluated for numerous metrics, described by following Equations, where, *tp*, *tn*, *fp*, and *fn* are "True Positive", "True Negative", "False Positive", and "False Negative" respectively (Alpaydin 2014), (Sokolova et al. 2006).

$$Acc = \frac{tp + tn}{tp + fp + fn + tn}. \tag{14}$$

$$F1 = \frac{2 \times tp}{2 \times tp + fp + fn}. \tag{15}$$

$$Pe = \frac{(tp + fp) \times (tp + fn) + (tn + fp) \times (tn + fn)}{(tp + tn + fp + fn)^2}. \tag{16}$$

$$Kap = \frac{Acc - Pe}{1 - Pe}. \tag{17}$$

$$TPR = \frac{tp}{(tp + fn)}. \tag{18}$$

$$FPR = \frac{fp}{(fp + tn)}. \tag{19}$$

The AUC and Kappa are also calculated alongside the Acc, and F1 (Chawla et al. 2004), (Hall et al. 2011). The AUC is computed from the graph, obtained by tracing the "True Positive Rate" (TPR) versus the "False Positive Rate" (FPR).

3.2 Experimental findings and discussion

The MCOV technique is applied to extract the classifiable features from the emotional EEG signals. The classification is carried out by using the ensemble learning approach and by utilizing the 10-fold cross-validation strategy (Salzberg 1997). As a result, the EEG dataset was separated into 10 parts. In the training phase, nine data components were used, while the remaining one was used in the testing procedure. The Waikato Environment for Knowledge Analysis is used to implement all considered classification algorithms (Eibe et al. 2016).

The findings of this experimentation show that the employment of the MCOV based features extraction with ensemble classification models achieved a high classification accuracy. Tables 1, 2, 3, 4, 5 and 6 describe the classifiers' performance for the considered problem. It is clear that all the classification techniques did pretty well in terms of the Acc, AUC, F1, and Kappa.

In the following parts the C1, C2 and C3 are respectively presenting the negative, neutral and positive emotions.

Table 1 summarizes different evaluation measures for considered classifiers. In this study the well-known classifiers, frequently used in literature for the categorization of EEG signals, are considered. These are namely, the SVM, k-NN, Random Forest (RF), Naïve Bayes Tree (NBTree), CART, C4.5, REPTree, and Random Ttree (RT).

The Puk kernel with an optimal value of C = 100 is used in the SVM implementation. The highest performance is attained with a k = 5 in the k-NN implementation. Here 'k' is presenting the number of neighbors. The judgment is determined on the basis of the Euclidian distance. The maximum performance in the implementation of RF is reached with a tree size of 100. The default settings render the maximum performances for the case of NBTree, CART, C4.5, REPTree and Random Tree.

For negative emotion category, the highest accuracy score of 91.70% is achieved with the RF. The SVM and CART are the second and third best performers with 87.60% and

Table 1 The performance of single classifiers

	Accuracy				F-Measure	AUC	Kappa
	Negative	Neutral	Positive	Average	Average	Average	Average
SVM	0.876	0.9	0.89	0.889	0.889	0.942	0.833
k-NN	0.795	0.627	0.685	0.702	0.703	0.777	0.5535
RF	0.917	0.805	0.913	0.878	0.878	0.971	0.8175
NBTree	0.794	0.686	0.773	0.751	0.751	0.853	0.6265
CART	0.859	0.723	0.841	0.808	0.807	0.881	0.7115
C4.5	0.855	0.748	0.824	0.809	0.809	0.868	0.7135
REPTree	0.802	0.701	0.81	0.771	0.771	0.88	0.6565
RT	0.785	0.65	0.752	0.729	0.729	0.797	0.5935

Table 2 The performance of Bagging based ensemble classifiers

	Accuracy				F-Measure	AUC	Kappa
	Negative	Neutral	Positive	Average	Average	Average	Average
SVM	0.917	0.927	0.917	0.92	0.921	0.986	0.8805
k-NN	0.992	0.981	0.971	0.981	0.981	0.999	0.972
RF	0.997	0.99	1	0.996	0.996	1	0.9935
NBTree	0.996	0.99	0.993	0.993	0.993	1	0.9895
CART	0.987	0.965	0.985	0.979	0.979	0.999	0.9685
C.4.5	0.988	0.975	0.995	0.986	0.986	0.999	0.979
REPTree	0.968	0.932	0.969	0.956	0.956	0.995	0.9345
RT	0.999	0.991	0.996	0.995	0.995	1	0.993

Table 3 The performance of Adaboost based ensemble classifiers

	Accuracy				F-Measure	AUC	Kappa
	Negative	Neutral	Positive	Average	Average	Average	Average
SVM	0.888	0.911	0.893	0.897	0.898	0.974	0.846
k-NN	0.795	0.627	0.685	0.702	0.703	0.777	0.5535
RF	0.909	0.817	0.911	0.879	0.879	0.97	0.8185
NBTree	0.854	0.769	0.843	0.822	0.822	0.94	0.733
CART	0.906	0.875	0.912	0.898	0.898	0.975	0.8465
C4.5	0.932	0.857	0.895	0.895	0.895	0.976	0.842
REPTree	0.889	0.842	0.887	0.873	0.873	0.969	0.809
RT	0.795	0.677	0.752	0.741	0.741	0.742	0.612

Table 4 The performance of Multiboost based ensemble classifiers

	Accuracy				F-Measure	AUC	Kappa
	Negative	Neutral	Positive	Average	Average	Average	Average
SVM	0.881	0.902	0.9	0.894	0.895	0.976	0.8415
k-NN	0.795	0.627	0.685	0.702	0.703	0.777	0.5535
RF	0.909	0.817	0.911	0.879	0.879	0.97	0.8185
NBTree	0.884	0.763	0.869	0.839	0.838	0.947	0.758
CART	0.903	0.831	0.901	0.878	0.883	0.97	0.8175
C4.5	0.92	0.822	0.908	0.883	0.883	0.97	0.825
REPTree	0.882	0.817	0.892	0.864	0.864	0.964	0.7955
RT	0.795	0.677	0.752	0.741	0.742	0.806	0.612

Table 5 The performance of RMS based ensemble classifiers

	Accuracy				F-Measure	AUC	Kappa
	Negative	Neutral	Positive	Average	Average	Average	Average
SVM	0.892	0.915	0.849	0.885	0.886	0.969	0.828
k-NN	0.846	0.716	0.745	0.769	0.769	0.906	0.6535
RF	0.904	0.777	0.884	0.855	0.854	0.963	0.7825
NBTree	0.88	0.734	0.86	0.825	0.824	0.943	0.737
CART	0.88	0.767	0.862	0.836	0.836	0.949	0.7545
C4.5	0.908	0.81	0.881	0.866	0.866	0.96	0.7995
REPTree	0.891	0.744	0.876	0.837	0.836	0.949	0.7555
RT	0.911	0.757	0.834	0.834	0.833	0.944	0.751

Table 6 The performance of RoF based ensemble classifiers

	Accuracy				F-Measure	AUC	Kappa
	Negative	Neutral	Positive	Average	Average	Average	Average
SVM	0.935	0.617	0.671	0.741	0.741	0.901	0.6115
k-NN	0.905	0.815	0.813	0.844	0.844	0.95	0.7665
RF	0.952	0.875	0.915	0.914	0.914	0.985	0.871
NBTree	0.943	0.855	0.913	0.904	0.903	0.979	0.8555
CART	0.952	0.879	0.912	0.914	0.914	0.984	0.8715
C4.5	0.956	0.893	0.921	0.923	0.923	0.986	0.885
REPTree	0.956	0.875	0.919	0.917	0.916	0.984	0.875
RT	0.937	0.842	0.843	0.874	0.874	0.968	0.811

85.90% accuracy scores respectively. The least accuracy score of 78.50% is obtained with the "Random Tree" (RT).

For neutral emotion class, the highest accuracy of 90.00% is secured by the SVM. The RF and C4.5 attain the second and third highest accuracies of 80.50% and 74.80% respectively. The k-NN attains the least accuracy score of 62.70%.

For positive emotion, the highest accuracy of 91.30% is obtained with the RF. The second and third highest accuracies of 89.00% and 84.10% are respectively achieved with the SVM and CART. The lowest accuracy score of 68.50% is achieved with the k-NN.

The highest average accuracy, for all three intended classes of 88.90% is achieved with the SVM. The RF follows with 87.80% accuracy. The third highest accuracy of 80.90% is achieved with the C4.5. The k-NN achieves the lowest accuracy score of 70.20%.

The highest and second highest average F-measure values of 88.90% and 87.80% are respectively obtained by the SVM and RF. The third highest F-measure of 80.90% is secured by the C4.5. The k-NN accomplishes the lowest F-measure of 70.30%.

The highest average AUC value of 97.10% is accomplished with the RF. The second and third highest AUC values of 94.20% and 88.10% are respectively obtained with the SVM and CART. The lowest AUC of 77.70% is achieved by the k-NN.

The highest and second highest average Kappa values of 83.30% and 81.80% are respectively obtained by the SVM and RF. The third highest Kappa of 71.40% is secured by the C4.5. The k-NN achieves the lowest Kappa of 55.40%.

Table 2 summarizes different evaluation measures for the Bagging based ensemble classifiers. For negative emotion category, the highest accuracy score of 99.90% is achieved with the Bagging-RT. The Bagging-RF secured the second highest accuracy of 99.70%. The least accuracy score of 91.70% is obtained with the Bagging-SVM.

For neutral emotion class, the highest accuracy of 99.10% is secured by the Bagging-RT. The Bagging-RF and Bagging-NBTree realized the second highest accuracy of 99.00%. The lowest accuracy of 92.70% is obtained with Bagging-SVM.

For positive emotion, the highest accuracy of 100.00% is obtained with the Bagging-RF. The second highest accuracy of 99.60% is achieved with the Bagging-RT. The lowest accuracy score of 91.70% is achieved with the Bagging-SVM.

The highest average accuracy of 99.60% is achieved with the Bagging-RF. The Bagging-RT follows with 99.50%

accuracy. The Bagging-SVM realizes the lowest accuracy score of 92.00%.

The highest and second highest average F-measure values of 99.60% and 99.50% are respectively obtained by the Bagging-RF and Bagging-RT. The Bagging-SVM realizes the lowest F-measure of 92.10%.

The highest average AUC value of 100.00% is accomplished with the Bagging-RF, Bagging-RT and Bagging-NBTree. The second highest AUC value of 99.90% is obtained with the Bagging-CART, Bagging-C4.5 and Bagging-k-NN. The lowest AUC of 98.60% is achieved by the Bagging-SVM.

The highest and second highest average Kappa values of 99.40% and 99.30% are respectively obtained by the Bagging-RF and Bagging-RT. The Bagging-SVM achieves the lowest Kappa of 88.10%.

Table 3 summarizes different evaluation measures for the Adaboost based ensemble classifiers. For negative emotion category, the highest accuracy score of 93.20% is achieved with the Adaboost-C4.5. The Adaboost-RF secured the second highest accuracy of 90.90%. The least accuracy score of 79.50% is obtained with the Adaboost-RT and Adaboost-k-NN.

For neutral emotion class, the highest accuracy of 91.10% is secured by the Adaboost-SVM. The Adaboost-CART achieved the second highest accuracy of 87.50%. The lowest accuracy of 62.70% is obtained with Adaboost-k-NN.

For positive emotion, the highest accuracy of 91.20% is obtained with the Adaboost-CART. The second highest accuracy of 91.10% is achieved with the Adaboost-RF. The lowest accuracy score of 68.50% is achieved with the Adaboost-k-NN.

The highest average accuracy of 89.80% is accomplished with the Adaboost-CART. The Adaboost-SVM follows with 89.70% accuracy. The Adaboost-k-NN accomplishes the lowest accuracy score of 70.20%. The highest average F-measure values of 89.80% is obtained by the Adaboost-CART and Adaboost-SVM. The second highest F-measure of 89.50% is secured by the Adaboost-C4.5. The Adaboost-k-NN accomplishes the lowest F-measure of 70.30%. The highest average AUC value of 97.60% is accomplished with the Adaboost-C4.5. The second highest AUC value of 97.50% is obtained with the Adaboost-CART. The lowest AUC of 74.20% is achieved by the Adaboost-RT. The highest and second highest average Kappa values of 84.70% and 84.60% are respectively obtained by the Adaboost-CART and Adaboost-SVM. The Adaboost-k-NN achieves the lowest Kappa of 55.40%.

Table 4 summarizes different evaluation measures for the Multiboost based ensemble classifiers. For negative emotion category, the highest accuracy score of 92.00% is achieved with the Multiboost-C4.5. The Multiboost-RF secured the second highest accuracy of 90.90%. The least accuracy score of 79.95% is obtained with the Multiboost-k-NN and Multiboost-RT.

For neutral emotion class, the highest accuracy of 90.20% is secured by the Multiboost-SVM. The Multiboost-CART achieved the second highest accuracy of 83.10%. The lowest accuracy of 62.70% is obtained with Multiboost-k-NN.

For positive emotion, the highest accuracy of 91.10% is obtained with the Multiboost-RF. The second highest accuracy of 90.80% is achieved with the Multiboost-C4.5. The lowest accuracy score of 68.50% is achieved with the Multiboost-k-NN.

The highest average accuracy of 89.40% is achieved with the Multiboost-SVM. The Multiboost-C4.5 follows with 88.30% accuracy. The Multiboost-k-NN accomplishes the lowest accuracy score of 70.20%. The highest average F-measure values of 89.50% is obtained by the Multiboost-SVM. The second highest F-measure of 88.30% is secured by the Multiboost-C4.5 and Multiboost-CART. The Multiboost-k-NN accomplishes the lowest F-measure of 70.30%. The highest average AUC value of 97.60% is achieved with the Multiboost-SVM. The second highest AUC value of 97.00% is obtained with the Multiboost-C4.5, Multiboost-CART and Multiboost-RF. The lowest AUC of 77.70% is achieved by the Multiboost-k-NN. The highest and second highest average Kappa values of 84.20% and 82.50% are respectively obtained by the Multiboost-SVM and Multiboost-C4.5. The Multiboost-k-NN achieves the lowest Kappa of 55.40%.

Table 5 summarizes different evaluation measures for the RSM based ensemble classifiers. For negative emotion category, the highest accuracy score of 91.10% is achieved with the RSM-RT. The RSM-C4.5 secured the second highest accuracy of 90.80%. The least accuracy score of 84.60% is obtained with the RSM-k-NN. For neutral emotion class, the highest accuracy of 91.50% is secured by the RSM-SVM. The RSM-C4.5 achieved the second highest accuracy of 81.00%. The lowest accuracy of 71.60% is obtained with RSM-k-NN. For positive emotion, the highest accuracy of 88.40% is obtained with the RSM-RF. The second highest accuracy of 88.10% is achieved with the RSM-C4.5. The lowest accuracy score of 74.50% is achieved with the RSM-k-NN.

The highest average accuracy of 88.50% is achieved with the RSM-SVM. The RSM-C4.5 follows with 86.60% accuracy. The RSM-k-NN accomplishes the lowest accuracy score of 76.90%. The highest average F-measure values of 88.60% is obtained by the RSM-SVM. The second highest F-measure of 86.60% is secured by the RSM-C4.5. The RSM-k-NN accomplishes the lowest F-measure of 76.90%. The highest average AUC value of 96.90% is accomplished with the RSM-SVM. The second highest AUC value of 96.30% is obtained with the RSM-NBTree. The lowest AUC of 90.60% is achieved by the RSM-k-NN. The highest and

second highest average Kappa values of 82.80% and 80.00% are respectively obtained by the RSM-SVM and RSM-C4.5. The RSM-k-NN achieves the lowest Kappa of 65.40%.

Table 6 summarizes different evaluation measures for the RoF based ensemble classifiers. For negative emotion category, the highest accuracy score of 95.60% is achieved with the RoF-REPTree and RoF-C4.5. The RoF-RF secured the second highest accuracy of 95.20%. The least accuracy score of 90.50% is obtained with the RoF-k-NN. For neutral emotion class, the highest accuracy of 89.30% is secured by the RoF-C4.5. The RoF-CART achieved the second highest accuracy of 87.90%. The lowest accuracy of 61.70% is obtained with RoF-SVM. For positive emotion, the highest accuracy of 92.10% is obtained with the RoF-C4.5. The second highest accuracy of 91.90% is achieved with the RoF-REPTree. The lowest accuracy score of 67.10% is achieved with the RoF-SVM.

The highest average accuracy of 92.30% is achieved with the RoF-C4.5. The RoF-REPTree follows with 91.70% accuracy. The RoF-SVM realizes the lowest accuracy score of 74.10%. The highest average F-measure values of 92.30% is obtained by the RoF-C4.5. The second highest F-measure of 91.60% is secured by the RoF-REPTree. The RoF-SVM realizes the lowest F-measure of 74.10%. The highest average AUC value of 98.60% is accomplished with the RoF-C4.5. The second highest AUC value of 98.50% is obtained with the RoF-RF. The lowest AUC of 90.10% is achieved by the RoF-SVM. The highest and second highest average Kappa values of 88.50% and 87.10% are respectively obtained by the RoF-C4.5 and RoF-REPTree. The RoF-SVM achieves the lowest Kappa of 61.20%.

This study provides a new emotion recognition framework based on the EEG signal processing. The MSPCA based denoising, MCOV based features extraction, and classification with ensemble models are the three primary steps of the proposed technique. The ensemble classifier with the highest performance is the Bagging with RF model. On other hand, the classifier with the poorest result is the k-NN (cf. Tables 1, 2, 3, 4, 5 and 6). The Acc of emotion recognition, F1, AUC, and Kappa are all used to evaluate the effectiveness of the proposed technique. Different well-known approaches are used for comparative reasons to demonstrate the success of the recommended methodology.

This paper presents one of the first applications to use a system that combines MSPCA, MCOV and ensemble classifiers for the EEG-based emotion identification. Several researchers have employed a variety of feature extraction and categorization methods.

Table 7 compares the accuracy of the proposed methodology with the state-of-the-art methods for processing the SEED dataset. It shows that Hwang et al., (2020) segmented the EEG signal and the STFT is computed for each segment. The band-pass filters are used to divide the spectrum in five sub-bands. The differential entropy features are extracted from each sub-band. In next step, the polar projection is applied to attain the topology preserving representation of the mined differential entropy features. Finally the CNN is used for processing the selected feature set to classify the emotions while securing 90.41% accuracy score.

In (Zhong et al. 2020), the authors have employed the local and global inter-channel relation via adjacency matrix for features extraction. The RGNN is sued for categorization. This model secures 94.24% accuracy score.

In (Wang et al. 2020), the authors have followed a similar sequence as presented in (Hwang et al., 2020) to obtain the sub-bands of short length segmented EEG signals. Onward, the EFDMs are generated by normalizing the obtained sub-band spectra. Finally the CNN is sued for the classification purpose. This model achieved 90.59% accuracy score.

In (Gao et al. 2020), the authors have followed the feature extraction scheme that was presented in (Hwang et al., 2020). Onward, the spatial-temporal information is extracted, from the differential entropy features, and the

Table 7 A comparison with previous studies

Study	Features Extraction	Classifier	Acc [%age]
(Hwang et al., 2020)	Topology-preserving differential entropy features	CNN	90.41
(Zhong et al. 2020)	Local and global inter-channel relation via Adjacency Matrix	RGNN	94.24
(Wang et al. 2020)	EFDMs	CNN	90.59
(Gao et al. 2020)	Differential entropy features + spatial-temporal information extraction	CDCN	90.63
(Wu et al. 2022)	Strength, clustering coefficient and Eigenvector centrality + statistical features of the eye movements.	DCCA	95.08
(Subasi et al. 2021)	MSPCA + TQWT	RoF with SVM	93.10
(Kumar and Molinas 2022)	DWT + DE	CNN	93.80
(Kamble and Sengupta 2021)	DWT + EMD + Statistical features from the IMFs	Ensemble Machine Learning	85.4%
The Proposed Framework	MSPCA + MCOV	Bagging with RF	99.60

classification is carried out via the CDCN model. This method secures 90.63% accuracy score.

In (Wu et al. 2022), the authors have derived five sub-bands of the intended EEG signals by using the band-pass filters. Onward, they extracted the strength, clustering coefficient and Eigenvector centrality from considered sub-bands. These features are fused with the statistically mined eye movement features. The selected feature set is then classified by the DCCA model while securing 95.08% accuracy score.

Subasi et al. have used a Symlets-4 filter based MSPCA for the noise diminishing of the EEG signals (Subasi et al. 2021). Onward, the TQWT is employed for features extraction. In next step, the statistical methods based dimension reduction is achieved. The selected feature set is processed by the RoF based ensemble classifiers while securing 93.10% accuracy score.

The authors in (Kumar and Molinas 2022) decomposed the EEG signals to five sub-bands. These sub-bands, obtained by using the DWT, are named as the delta, theta, alpha, beta, and gamma. Onward, the "Differential Entropy" (DE) based features are mined from the intended sub-bands. The extracted feature set is processed by the CNN for the categorization of emotions. This model secures 93.8% accuracy score for the SEED dataset.

The authors in (Kamble and Sengupta 2021) decomposed the EEG signals to four sub-bands. These sub-bands, obtained by using the DWT, are named as the theta, alpha, beta, and gamma. Onward, the EMD based IMFs are derived from the intended sub-bands. Afterward, statistical features are extracted from IMFs to prepare the feature set. The extracted feature set, obtained from the DREAMER dataset, are used to train the "Ensemble Machine Learning" (EML) model. A similar decomposition and feature extraction process is carried out for the SEED dataset. The performance of pretrained EML model is validated by processing the feature set, mined from the SEED dataset. This model secures 85.4% accuracy score for the SEED dataset.

The mobile healthcare is getting popular because of its attractive features such as cost effectiveness plus remote and continuous monitoring of concerned subjects. This approach is beneficial for remote areas particularly in the developing countries (Subasi et al. 2020). It is mainly based on the wireless wearables which are connected to the cloud health servers via the internet of medical things (IoMT) (Mian Qaisar and Hussain 2021). In this context, it is recommended to perform the pre-processing and feature extraction processes on the front-end module and realize the classification task on cloud server (Mian Qaisar and Subasi 2020). An effective realization can render an efficient front-end module with a reduced data dimension to handle and transmit. Moreover, the classification module remains easily configurable on the cloud (Mian Qaisar and Subasi 2020). In this scenario, the computational complexity of devised method is compared with the previously published works by the author (Subasi et al. 2021), (Tuncer et al. 2021b, p. 19). In this procedure the focus is made on the pre-processing and feature extraction stages, intended to be implemented on the front-end module. The classification, for all intended studies, is carried out by using ensemble machine learning algorithms. Therefore, the complexity of classification stage is assumed analogous for all cases and is not compared.

The signal acquisition, segmentation, pre-processing and feature extraction approach is similar for (Subasi et al. 2021) and (Tuncer et al. 2021b, p. 19) (cf. Figure 2).

Figure 2 shows that in both studies, the symlets-4 filter based MSPCA is used for pre-processing and the features are extracted using the TQWT, generating 18 sub-bands. However, the dimension reduction processes are different. In (Subasi et al. 2021), 6 statistical features are extracted from each sub-band. In (Tuncer et al. 2021b, p. 19) 540 features are extracted from each sub-band and these are fused with other 540 features extracted from the EEG instance waveform, resulting in 10,260 features per instance. These features are mined using a specifically suggested method, named as the LEDPatNet19. Onward the ReliefF and Chi2 (RFIChi2) feature selector is used for dimension reduction.

By comparing Figs. 1 and 2 it is evident that the signal acquisition, segmentation and pre-processing processes are similar in the suggested and considered counterparts. Therefore, complexities of the common processes are not compared. The feature extraction approach is different between the proposed method and the ones presented in (Subasi et al. 2021) and (Tuncer et al. 2021b, p. 19). Therefore, the complexity of this process is compared.

In the case of (Subasi et al. 2021) and (Tuncer et al. 2021b, p. 19), after pre-processing each conditioned instance is decomposed to 18 sub-bands using half-band filters banks. Let the scaling factors α and β are respectively for the low pass and high pass half-band filters. Then, the first level

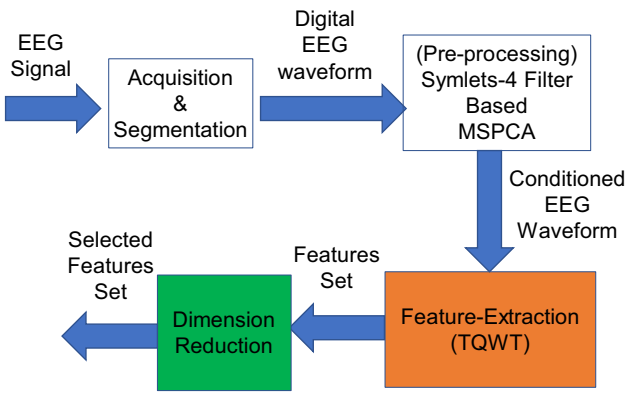


Fig. 2 The EEG pre-processing and feature extraction chain, used in the related works

decomposition stage receives EEG signal, sampled at F_S, and delivers detailed and approximation coefficients respectively sampled at $\beta \times F_S$ and $\alpha \times F_S$. The approximation coefficients are delivered to the second stage of decomposition and the process continues up to the eighteenth stage.

The Quality factor (Q), can be computed as: $Q = \frac{C_0}{B.W}$. Here, C_0 and $B.W$ are respectively the signal central frequency and bandwidth. The relationship between α, β, and Q is given by Eq. (20). Where, r is the redundancy factor. The range of β is $0 < \beta \leq 1$ and range of α is $0 < \alpha < 1$ (Selesnick 2011).

$$\beta = \frac{2}{Q+1} \; and \; \alpha = 1 - \frac{\beta}{r}. \tag{20}$$

Assume that the incoming instance is composed of N samples. The process of decomposition and scaling at each stage is achievable by zero-padding and FIR filtering (Selesnick 2011). This assembly can be offline designed. Hence, the online cost of TQWT based decomposition, C_{TQWT}, in terms of arithmetic operations can be expressed by Eq. (21) (Mian Qaisar and Hussain 2021). Where, M is the order of half-band finite impulse response (FIR) filters and $k+1$ is indexing the number of decomposition stage.

$$\begin{aligned} C_{TQWT} = & \sum_{k=0}^{17} \left[\underbrace{(M-1) \times \beta \times \alpha^k \times N}_{Additions} + \underbrace{M \times \beta \times \alpha^k \times N}_{Multiplications} \right] \\ & + \sum_{k=0}^{17} \left[\underbrace{(M-1) \times \alpha^{k+1} \times N}_{Additions} + \underbrace{M \times \alpha^{k+1} \times N}_{Multiplications} \right]. \end{aligned} \tag{21}$$

In the proposed method, the features are extracted using the MCOV. The process of approximating the AR model parameters is realized offline. During online processing, the N spectral coefficients are computed using the FFT. The cost of this operation is: $\underbrace{N \times log_2 N}_{Additions} + \underbrace{N \times log_2 N}_{Multiplications}$. Onward a 15th order AR model is used to approximate P, P_{MCOV} coefficients. Therefore, the computational cost C_{MCOV} is given by Eq. (22) for this case.

$$C_{MCOV} = \underbrace{\left(N \times log_2 N\right) + 15 \times P}_{Additions} + \underbrace{P + 15 \times P\left(N \times log_2 N\right)}_{Multiplications}. \tag{22}$$

In the case of (Subasi et al. 2021), (Tuncer et al. 2021b, p. 19), the TQWT with Q=2 and r=3 is used. It renders β =0.667 and α=0.778. For M=5 and N=1024 the C_{TQWT} is 23,670 additions and 32,963 multiplications.

For the devised approach for P=40 and N=1024 the C_{MCOV} is 10,840 additions and 10,880 multiplications. It shows that the gain in additions and multiplications, of MCOV over the TQWT, for the studied case is respectively 2.43 and 3.03 times. Additionally, in the case of (Subasi et al. 2021) and (Tuncer et al. 2021b, p. 19), the dimension reduction is attained by processing the TQWT delivered 18 sub-bands. It is realizable by performing extra addition, subtraction and multiplication operations (Mian Qaisar and Hussain 2021). However, in the proposed approach the dimension reduction is not required. It confirms that the MCOV based proposed feature extraction mechanism is effective in terms of computational cost over the TQWT based approach, used in the intended counterparts.

The suggested system outperforms in terms of the emotion recognition accuracy, as outlined in Table 7. Additionally, it is cost effective as compared to the previous works. When compared to other methods, these findings clearly indicate the efficiency of the proposed framework in the EEG-based emotion identification. Furthermore, the method outperformed the deep learning approaches in terms of recognition rate. The key merits of the proposed approach are listed below.

- All stages of the suggested technique have a simple mathematical basis. Therefore, it can lead towards an efficient realization as compared to the previous works and other deep learning-based approaches.
- The suggested MCOV-based features extraction approach renders an excellent classification performance. It confirms the success of proposed features extraction method and demonstrates that the extracted attributes are idiosyncratic. In this study the highest identification performance is attained by using the AR model with order of 15. The optimal model order is determined by trial and error. 40 PSD coefficients are utilized to present each instance to get better performance. The obtained PSD values are further processed by the classifiers.
- A framework for very accurate EEG-based emotion identification is provided.
- A comparison with state-of-the-art confirms the effectiveness of the recommended strategy.

4 Conclusion

This paper provides a unique MCOV and ensemble classifier based emotion recognition system by processing the EEG signals. The suggested system includes MSPCA based denoising, MCOV based feature extraction, and classification with ensemble learning. A benchmarking is made with state-of-the-art techniques. Utilizing the Bagging + Random Forest ensemble classifier, the suggested MCOV based system obtained 99.6% classification accuracy. In conclusion, this research presents a unique and accurate EEG data processing approach for emotion identification. The suggested

approach is lightweight as it is using the simple mathematical models. For the studied case the computational gain of 2.43 and 3.03 times is secured respectively in terms of additions and multiplications over the previous counterparts. It confirms the effectiveness of the devised approach. The performance of devised approach is only presented for the intended dataset. It can limit the generalizability of findings. Therefore, in future other potential EEG emotion datasets with multiple scenarios and long-term records will be considered for confirming the robustness and reliability of the proposed MCOV-based ensemble emotion detection method. Additionally, a bigger variety of emotions can be considered and categorized. Moreover, the subject dependent validation can be explored with explainability. Furthermore, the proposed method can be used to process the EEG signals for other applications like the stress identification and epilepsy detection. As well as other biological signals like the electrocardiogram (ECG) and electromyogram (EMG) can be examined and categorized using this approach. Additionally, the feasibility of incorporating the other features mining techniques like the empirical mode decomposition and autoregressive analysis will be investigated. Examining the performance of deep learning approaches for automated feature mining and categorization is another prospect.

Funding This work was supported by Effat University with the Decision Number of UC#9/29 April.2020/7.1–22(2)5, Jeddah, Saudi Arabia.

Data availability The dataset used in this paper is publicly available at: https://bcmi.sjtu.edu.cn/home/seed/.

Declarations

Conflict of interest Authors declare no conflict of interest. The second author (S. M. Qaisar) has recently moved to the CESI LINEACT, France and certain parts of the manuscript were written or revised while he was with the Effat University.

Ethical approval This article does not contain any studies with human participants or animals performed by any of the authors.

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